Topic 4 – Extracting metals and equilibria

Obtaining and using metals

Students should:		Maths skills
4.1	Deduce the relative reactivity of some metals, by their reactions with water, acids and salt solutions	
4.2	**Explain displacement reactions as redox reactions, in terms of gain or loss of electrons**	
4.3	Explain the reactivity series of metals (potassium, sodium, calcium, magnesium, aluminium, (carbon), zinc, iron, (hydrogen), copper, silver, gold) in terms of the reactivity of the metals with water and dilute acids and that these reactions show the relative tendency of metal atoms to form cations	
4.4	Recall that: a most metals are extracted from ores found in the Earth’s crust b unreactive metals are found in the Earth’s crust as the uncombined elements	
4.5	Explain oxidation as the gain of oxygen and reduction as the loss of oxygen	
4.6	Recall that the extraction of metals involves reduction of ores	
4.7	Explain why the method used to extract a metal from its ore is related to its position in the reactivity series and the cost of the extraction process, illustrated by a heating with carbon (including iron) b electrolysis (including aluminium) (knowledge of the blast furnace is not required)	
4.8	**Evaluate alternative biological methods of metal extraction (bacterial and phytoextraction)**	
4.9	Explain how a metal’s relative resistance to oxidation is related to its position in the reactivity series	
4.10	Evaluate the advantages of recycling metals, including economic implications and how recycling can preserve both the environment and the supply of valuable raw materials	
4.11	Describe that a life-cycle assessment for a product involves consideration of the effect on the environment of obtaining the raw materials, manufacturing the product, using the product and disposing of the product when it is no longer useful	
4.12	Evaluate data from a life cycle assessment of a product	

Suggested practicals

- Investigate methods for extracting metals from their ores.
- Investigate simple oxidation and reduction reactions, such as burning elements in oxygen or competition reactions between metals and metal oxides.

Reversible reactions and equilibria

Students should:		Maths skills
4.13	Recall that chemical reactions are reversible, the use of the symbol ⇌ in equations and that the direction of some reversible reactions can be altered by changing the reaction conditions	
4.14	Explain what is meant by dynamic equilibrium	
4.15	Describe the formation of ammonia as a reversible reaction between nitrogen (extracted from the air) and hydrogen (obtained from natural gas) and that it can reach a dynamic equilibrium	
4.16	Recall the conditions for the Haber process as: a temperature 450 °C b pressure 200 atmospheres c iron catalyst	
4.17	**Predict how the position of a dynamic equilibrium is affected by changes in:** **a temperature** **b pressure** **c concentration**	

Suggested practicals

- Investigate simple reversible reactions, such as the decomposition of ammonium chloride.